



CSCI 295: SpecTop- Full-Stack Web Development (Spring 2025)

Class time: Tuesday/Thursday, 8:00 – 9:50am (In-Person)

Location: Paine 212- Computer/Data Science classroom

Instructor

Babafemi Sorinolu

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- **Office:** Paine 209B; 585.567.9259
- **Office hours:** Tuesday/Thursday, 2:30-4:30pm (Feel free to email me to schedule a virtual meeting.)

Moodle Page

<https://moodle.houghton.edu/course/view.php?id=28901>

Course Description

(4 credit hours) - Covers the foundational technologies of HTML, CSS, and JavaScript, and how to create responsive, interactive websites using React and NodeJS technologies.

Prerequisite: NA.

Optional Textbooks

Full-Stack React Projects, 2nd Edition by Shama Hoque

Additional Textbook

NA

Required Software

Vs code, Node, MongoDB, MySQL Workbench

Long-Range Goals:

1. Equip students with the fundamental skills needed to build basic, interactive, and responsive web pages using HTML, CSS, and JavaScript.
2. Introduce students to the full development cycle, from designing a user interface (UI) to writing clean code, testing functionality, and ultimately deploying a live web page.
3. Lay the groundwork for students to transition into more advanced web development topics, such as back-end development with databases, server-side programming, and frameworks like Node.js

Course Objectives: After successful completion of this course, students will be able to:

- Understand the fundamentals of web development, including how web browsers, servers, and HTTP protocols interact to deliver web pages, and recognize the roles of HTML, CSS, and JavaScript in building a website
- Create well-structured web pages using HTML and style web pages with CSS
- Add interactivity to web pages using JavaScript, including handling events, manipulating the DOM to dynamically alter content, and implementing basic JavaScript logic for tasks like form validation and creating interactive features.
- Collaborate using Git and GitHub
- Deploy web pages to the internet

Activities & Time Commitment: For a four-credit course, you should expect to commit at least 12 hours per week. Although not every week will be the same, these are typical weekly times:

- **Weekly Lectures (4 hours):** Lectures will cover theoretical concepts, such as web development basics, HTML, CSS, JavaScript syntax, and web design principles. These sessions will be interactive, with live coding examples and Q&A.
- **Hands-On Lab Sessions (3 hours):** In these lab sessions, students will work on coding exercises with immediate feedback from the instructor or teaching assistant. This allows students to apply what they've learned in a practical, guided environment.
- **Homework (2 hours):** Students will complete small assignments that reinforce the concepts learned in class. These will involve tasks like building a web page using HTML/CSS, implementing a JavaScript feature (like form validation), or debugging code. **Work is due at the time it is assigned**, whether you are in class or not.
- **Group Projects (3 hours):** Students will work on a collaborative group project where they will apply HTML, CSS, and JavaScript to build a multi-page, interactive website. This project will require teamwork, communication, and version control using Git/GitHub.

Evaluation and Grading: The grading and grading scale will be based on the following tables:

- **Grading:** The evaluation will be based on the following criteria and weights.

Criteria	Weights (%)
Homework	40
Midterm Project	20
Final Project	40

- **Grading Scale**

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
%	100-93	92-90	89-87	86-83	82-80	79-77	76-73	72-70	69-67	66-63	62-60	59-

All final grades will be rounded up to the nearest whole number when assigning letter grades.

Attendance: Let me know (by email to babafemi.sorinolu@houghton.edu with the subject "CSCI 218 Attendance") if and why you will be missing a class. In general, this should be done before the class you will miss.

Exams: There will be one midterm exam in the class and one cumulative final exam. Make-up exams will only be given in extenuating circumstances; prior arrangements must be made if possible.

Tentative dates for each exam are as follows:

- Midterm Project: Tuesday, March 11th
- Final Project due date: Tuesday, May 6th, 8:00–10:00am.

Academic Honesty: Honesty is the foundation on which all intellectual endeavors rest. To use the ideas of others without acknowledging the authors of those ideas belies the nature and purpose of academic life. At Houghton, where we strive to live out Christian calling and commitment, personal integrity, including academic honesty, should be the hallmark of all our work and relationships. Students are expected to exhibit extreme care relative to personal honesty in all academic work, including in-class and out-of-class learning experiences, such as exams, quizzes, journals, papers, and research projects. Please refer to the catalog for the details of this policy.

Self-Reporting of Disabilities: If you have an academic or physical disability that requires accommodations, it is up to you to self-report any such disability to the office of Academic Support and Accessibility Services in the Center for Student Success located on the 1st floor of Chamberlain. (585-567-9622) With appropriate documentation, you will be afforded

the necessary accommodation. For more information about the Center for Academic Success and Advising, go to:- <https://www.houghton.edu/current-students/center-for-student-success/academic-support-and-accessibility-services/>

Tentative Schedule: Below is a tentative calendar for the topics we will cover during each class period. These topics correspond to the chapters in our textbook.

Week	Class Date	Lecture Topics
1	1/14 - 1/16	Basic Built-in Terminal Commands Creating a Repository on GitHub
2	1/21 - 1/23	Introduction to HTML
3	1/28 - 1/30	HTML Elements Overview HTML Attributes
4	2/4 - 2/6	CSS Basics & Styling
5	2/11 - 2/13	Advanced HTML & CSS (Forms, Layouts, Animations)
6	2/18 - 2/20	Introduction to JavaScript (JS)
7	2/25 - 2/27	Introduction to JavaScript (JS)
	3/4 - 3/6	Mid-semester break
8	3/11 - 3/13	Advanced JavaScript Concepts
9	3/18 - 3/20	Introduction to Web Servers and Node.js
10	3/25 - 3/27	Building RESTful APIs with Node.js and Express
11	4/1 - 4/3	Database Fundamentals - SQL and NoSQL
12	4/8 - 4/10	Authentication and Authorization
13	4/15	Front-End Frameworks - React.js
	4/17 - 4/22	Easter break
14	4/22 - 4/24	Building a React Application
15	4/29 - 5/1	Review
16	5/6	Final Exams